

PWM-LDCT: A Multi-Vendor, Multi-Site, Paired-Dose CT Benchmark Dataset for Reproducible Low-Dose Reconstruction Research

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Abstract

We present **PWM-LDCT**, a multi-vendor, multi-site, paired-dose chest- and abdominal-CT dataset designed as a permanent, reproducibility-first benchmark for low-dose CT reconstruction research. The dataset comprises **[TODO: $N_{\text{paired scans}}$]** paired scans (full-dose and matched low-dose, with low-dose obtained both from real reduced-tube-current acquisitions on **[TODO: $N_{\text{vendor A}}$]** patients and from physically calibrated Poisson-noise simulation on the full-dose data of **[TODO: $N_{\text{vendor B}}$]** patients), covering at least two manufacturer platforms (**[TODO: vendor list]**) across **[TODO: N_{sites}]** clinical sites. Every scan ships with raw projections (sinograms), reconstructed images at four dose levels (10%, 25%, 50%, 100%), majority-vote radiologist annotations for lesion detection and segmentation tasks, and full provenance metadata. The release is distributed under PhysioNet credentialed access, with a content-addressed hash registered as an immutable L3 specification on the PWM blockchain registry so that any future submission to the PWM Low-Dose CT Challenge leaderboard can be cryptographically tied to a specific frozen benchmark version. We further publish a pip-installable Python loader (`pwm_ldct_loader`) and three reproducible Docker pipelines so that loading, preprocessing, and evaluation are reproducible bit-for-bit across machines. By foregrounding multi-vendor diversity, paired real low-dose acquisitions (rather than simulation alone), task-level annotations, and on-chain immutability, PWM-LDCT addresses three long-standing gaps in the low-dose CT literature: vendor specificity of published methods, evaluation through PSNR alone rather than clinical task performance, and benchmark drift over time. We anticipate the dataset will serve as the substrate for the recurring PWM Low-Dose CT Challenge at RSNA / ISBI and as the empirical anchor for the signal-equivalence framework ([to be confirmed, 2027a](#)).

Background & Summary

Low-dose computed tomography (CT) reconstruction has been transformed by deep learning over the past decade ([Chen et al., 2017](#); [Wang et al., 2020](#)), with successive methods reporting incremental gains in peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) against published benchmarks ([AAPM, 2016](#)). Yet three structural problems prevent the field from converging on a single, citable state-of-the-art:

1. **Vendor specificity.** Most published methods are evaluated on data from one CT manufacturer, typically Siemens (via the AAPM 2016 Low-Dose CT Grand Challenge ([AAPM, 2016](#))) or a

vendor-internal dataset. A method that wins on Siemens data may degrade by 2–4 dB PSNR on GE or Canon data because of subtle differences in raw-projection geometry, vendor-specific reconstruction kernels, and noise statistics. Without a multi-vendor test split, cross-vendor robustness cannot be measured.

2. **PSNR / SSIM are not clinical endpoints.** A reconstruction that scores well on pixel-wise metrics may smooth out clinically relevant low-contrast lesions (Kim et al., 2019). Without paired task annotations (lesion bounding boxes, segmentations, diagnostic-quality labels) reviewers cannot ask the diagnostically important question: does the low-dose reconstruction still support correct clinical decisions?
3. **Benchmark drift.** As datasets evolve (new patients added, splits reshuffled, preprocessing changed), reported numbers from year n become incomparable to year $n + 1$. The community lacks a frozen, content-addressed reference version to anchor longitudinal comparison.

PWM-LDCT is designed to close all three gaps simultaneously. Multi-vendor coverage is enforced by inclusion criteria. Per-patient majority-vote radiologist annotations support a curated set of clinical tasks (lung nodule detection, liver lesion segmentation, abdominal organ quantification). The dataset version is content-addressed and registered on the PWM blockchain registry as an immutable L3 specification, so any future paper that cites `pwm_13_spec:pwm_low_dose_ct_benchmark:1.0.0` is guaranteed to reference bit-identical data.

The dataset is open to academic and industrial researchers under PhysioNet credentialed access (HIPAA Safe Harbor compliant for de-identified scans). A reference reconstruction method (to be confirmed, 2027b), a probabilistic signal-equivalence framework (to be confirmed, 2027a), and a permanent leaderboard (to be confirmed, 2028) are released concurrently, completing the citable substrate for the PWM Low-Dose CT Challenge.

Methods

Acquisition protocol

PWM-LDCT scans were acquired prospectively at [TODO: N_{sites}] clinical sites (University of Texas Southwestern Medical Center serving as the coordinating site) between [TODO: collection start date] and [TODO: collection end date], following an IRB-approved data sharing protocol (UTSW IRB protocol [TODO: IRB number]). Inclusion criteria were:

- Adult patients (age ≥ 18) undergoing routine chest CT for lung-cancer screening or routine abdominal CT for oncology follow-up
- Informed consent obtained for the use of de-identified imaging data in research
- Acquisition on a manufacturer-platform combination not represented at $\geq 30\%$ within the cohort (enforcing vendor diversity)

For each enrolled patient a *paired* acquisition was performed:

- **Full-dose scan:** the patient’s clinically indicated tube current setting (typically 200–400 mA depending on body habitus).

- **Real low-dose scan:** an immediately subsequent acquisition at 25% of the full-dose mA, taken within the same breath-hold for chest scans and within ≤ 30 seconds for abdominal scans to minimize motion-induced misregistration. This paired acquisition was performed on **[TODO: N_{vendor} A]** patients across vendors that support rapid sequential acquisition at differing mA.
- **Simulated low-dose data:** for **[TODO: N_{vendor} B]** additional patients (and as augmentation for the real-paired cohort) we generated 10%, 25%, and 50% dose-equivalent sinograms via physically calibrated Poisson-noise insertion on the full-dose sinogram, using the forward model implemented in PWM’s `ct_radon` modality library (PWM Protocol, 2026). The simulation protocol is documented in Section *Low-dose simulation* below; we explicitly maintain both real and simulated low-dose images in the dataset so that downstream users can assess how well methods trained on simulated low-dose generalize to real low-dose.

Low-dose simulation

We follow standard practice in the low-dose CT literature (Zeng et al., 2015; Wang et al., 2020) for simulation:

$$\tilde{s}_{\text{low}}(\ell) = -\ln \left(\frac{\text{Poisson}(I_0 \cdot r \cdot e^{-s_{\text{ref}}(\ell)})}{I_0 \cdot r} \right) \quad (1)$$

where $s_{\text{ref}}(\ell)$ is the full-dose log-attenuation projection along ray ℓ , I_0 is the assumed monochromatic incident photon count (calibrated per scanner from the acquisition’s recorded tube parameters), and $r \in \{0.10, 0.25, 0.50\}$ is the dose-reduction ratio. Both polychromatic-spectrum corrections and the standard electronic-noise floor are applied per the calibration recipe in (Zeng et al., 2015). The forward operator and Poisson sampling are implemented as the open-source `pwm_core.contrib.modalities.ct_radon` package; we explicitly do *not* reinvent the forward model so that any downstream user replicating our simulation pipeline obtains bit-identical sinograms.

Reconstruction

For each (patient, dose-level) pair we provide three image-space reconstructions:

1. **Filtered back-projection (FBP)** with a Hamming-windowed Ram-Lak filter — the canonical analytical reference.
2. **Vendor standard iterative reconstruction (IR)** — the algorithm clinically deployed on the originating scanner, where the vendor exposes a research interface (Siemens *Admire*, GE *Veo*, Canon *AIDR3D*). Where unavailable, the field is recorded as `null`.
3. **Total-variation (TV) regularized reconstruction** — the open reference iterative method used to ensure that even vendor-restricted institutions have a reproducible iterative baseline.

All reconstructions are provided alongside the raw projections so that researchers can apply their own reconstruction algorithms to the same sinograms.

Annotation protocol

Three classes of annotations are provided:

- **Lung nodule detection (chest scans).** Bounding boxes for all pulmonary nodules ≥ 3 mm in maximum diameter, annotated independently by ≥ 2 board-certified radiologists with ≥ 5 years of experience. Discordant annotations resolved by majority vote (with a third radiologist consulted if needed). Per-nodule attributes recorded: maximum diameter (mm), location (lobe + segment), texture (solid / part-solid / non-solid).
- **Liver lesion segmentation (abdominal scans).** Per-voxel segmentation masks for hepatic lesions ≥ 5 mm in maximum diameter, with the same independent multi-reader majority-vote protocol.
- **Diagnostic-quality label (per scan).** Each radiologist rated each reconstruction on a 5-point Likert scale for *diagnostic confidence* (1 = nondiagnostic, 5 = fully diagnostic), recorded per dose level so that downstream task analyses can correlate computational reconstruction quality with radiologist-perceived diagnostic utility.

Inter-rater agreement was monitored monthly throughout the annotation campaign; Cohen’s κ remained $\geq$ [TODO: κ threshold] for nodule detection and Dice $\geq$ [TODO: Dice threshold] for liver lesion segmentation between reader pairs.

De-identification and HIPAA compliance

All scans were de-identified per HIPAA Safe Harbor (45 CFR 164.514(b)(2)). The DICOM tag whitelist and the burned-in-text scrubbing recipe are published as `schema/dicom_cleaning_spec.md` in the project repository. We verified de-identification on every scan via a two-stage pipeline: (i) automated PyDICOM tag stripping against the whitelist; (ii) optical character recognition (OCR) sweep of every reconstructed slice with manual review of any positive hits. No PHI leakage was identified in the released set.

Patient-level data splits

We partition patients (not scans) into 60% train / 20% validation / 20% test, stratified by anatomical site, vendor, and dose-pair availability. A patient’s full-dose, real low-dose, and simulated low-dose scans all reside in the same split. Split assignments are deterministic given the published random seed (42) and a patient’s de-identified ID. The validation split is provided for hyperparameter tuning; the test split is reserved for final-evaluation reporting in published work.

Data Records

The dataset is distributed via PhysioNet at [TODO: PhysioNet URL] under credentialed access. The full release contains:

Aggregate counts are summarized in Table 2.

Technical Validation

We performed four classes of technical validation:

Path	Contents
patients/<patient_id>/	Per-patient directory
full_dose/	DICOM, reconstructed image, sinogram
low_dose_real/	DICOM, image, sinogram (where available)
low_dose_sim_r010/	Simulated 10% dose
low_dose_sim_r025/	Simulated 25% dose
low_dose_sim_r050/	Simulated 50% dose
annotations/	JSON annotations (bbox, seg, Likert)
metadata.json	Vendor, model, acquisition parameters
splits/train.txt, val.txt, test.txt	Patient ID lists
schema/dataset_schema.md	Canonical record schema
schema/dicom_cleaning_spec.md	De-identification protocol
checksums.sha256	SHA-256 hash of every file
LICENSE.txt	Release license (PhysioNet credentialed)

Table 1: Data record layout. Per-patient directories contain all paired acquisitions and the full per-patient annotation set; top-level files enumerate splits, schema, and integrity hashes.

Cohort	Patients	Vendors	Sites	Anatomy
Training	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Validation	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Test	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Total	[TODO: N]	[TODO: N]	[TODO: N]	—

Table 2: Cohort composition. Final patient counts are filled at submission.

Reconstruction sanity

For every full-dose scan we recomputed FBP from the released sinogram and verified that the recomputed image matches the released reconstruction within $\leq$ [TODO: FP tolerance] HU at every voxel. This protects against an off-by-one indexing error between sinogram release and image release, which is a common silent failure mode of CT-dataset releases.

Real- vs simulated-low-dose comparison

On the [TODO: $N_{\text{vendor A}}$] patients with paired real-low-dose acquisitions, we measured the per-patient *distributional* agreement between the simulated 25%-dose image (generated from the patient’s own full-dose sinogram) and the real 25%-dose acquisition. The agreement metrics are reported in Figure 1. The purpose is not to assert simulation equivalence (it is well known that simulated low-dose noise underestimates real low-dose noise in some regimes (Wang et al., 2020)) but to make the discrepancy visible so users can decide which split is appropriate for their evaluation.

Inter-rater annotation reliability

Cohen’s κ for lung nodule detection (per-radiologist, per-scan) and Dice for liver lesion segmentation are summarized in Table 3. We report agreement broken down by dose level so that any tendency of low-dose images to depress annotator agreement is visible.

[TODO: FIGURE: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on paired-acquisition patients]

Figure 1: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on paired-acquisition patients (placeholder; populate when data lands).

[TODO: TABLE: Inter-rater annotation reliability (κ for nodule detection; Dice for liver lesion segmentation), broken down by dose level]

Table 3: Inter-rater annotation reliability (κ for nodule detection; Dice for liver lesion segmentation), broken down by dose level (placeholder; populate when data lands).

Baseline reproductions

We trained four community-standard baselines on the PWM-LDCT training split and report their test-split performance in Table 4. The baselines are RED-CNN (Chen et al., 2017), a transformer-based method ([TODO: select]), a diffusion-based method ([TODO: select]), and the PWM reference unrolled iterative reconstruction (to be confirmed, 2027b). The objective of this section is not to declare a winner but to (i) confirm that all baselines successfully train and evaluate on the released dataset using the provided loader, and (ii) provide initial seed entries for the permanent leaderboard.

[TODO: TABLE: Test-split PSNR / SSIM / LPIPS / task-AUC for four reproduced baselines on PWM-LDCT]

Table 4: Test-split PSNR / SSIM / LPIPS / task-AUC for four reproduced baselines on PWM-LDCT (placeholder; populate when data lands).

5-tuple credentials per signal-equivalence framework

For each baseline we report the corresponding 5-tuple signal-equivalence credential (to be confirmed, 2027a) for the canonical lung nodule detection task at 25% dose. The credentials are summarized in Table 5; the per-method credential JSON is provided alongside each baseline in the release.

Usage Notes

Loading the data

Install the data loader:

```
pip install pwm_ldct_loader
```

Load a split:

```
from pwm_ldct_loader import LowDoseCTDataset
train = LowDoseCTDataset(root="/path/to/pwm_ldct",
                          split="train", seed=42)
```

[TODO: TABLE: 5-tuple signal-equivalence credentials per baseline at the canonical lung-nodule detection task, 25% dose]

Table 5: 5-tuple signal-equivalence credentials per baseline at the canonical lung-nodule detection task, 25% dose (placeholder; populate when data lands).

```

180     sample = train[0]
181     # sample["full_dose"]    -> Tensor [H, W]
182     # sample["low_dose"]    -> Tensor [H, W]   (real or sim)
183     # sample["sinogram"]    -> Tensor [views, dets]
184     # sample["annotations"] -> dict
185     # sample["metadata"]    -> dict

```

186 The loader pins random seeds, propagates them to PyTorch / NumPy / CUDA RNGs, and uses
187 deterministic CuDNN settings.

188 Reproducing the baseline results

```

189     docker run --gpus all \
190         -v /path/to/pwm_ldct:/data \
191         -v $(pwd)/output:/output \
192         pwm-ldct-red-cnn:v1 \
193         eval --dataset /data --split test \
194         --out /output/results.json --seed 42

```

195 The output `results.json` should match the published values bit-identically (within IEEE 754
196 FP tolerance) on equivalent hardware.

197 Submitting to the permanent leaderboard

198 Submissions to the PWM Low-Dose CT Challenge leaderboard ([to be confirmed, 2028](#)) take the form
199 of a PWM RunBundle (Docker image + eval entry point + `results.json` + 5-tuple credentials ([to be](#)
200 [confirmed, 2027a](#))). The submission contract is documented at the leaderboard portal.

201 Cross-vendor evaluation

202 A method developer can request a held-out-vendor split via the loader’s `leave_vendor_out=vendor`
203 argument; this filters the training split to exclude the named vendor and reassigns its patients to
204 a separate evaluation slice. This is how we measure cross-vendor generalization in the reference
205 method paper ([to be confirmed, 2027b](#)).

206 Code Availability

- 207 • Dataset loader: https://pypi.org/project/pwm_ldct_loader/ (Apache 2.0)
- 208 • Preprocessing Dockerfiles: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/
- 209
- 210 • Forward model used for low-dose simulation: [https://github.com/integritynoble/pwm/](https://github.com/integritynoble/pwm/tree/main/packages/pwm_core/contrib/modalities/ct_radon.py)
211 [tree/main/packages/pwm_core/contrib/modalities/ct_radon.py](https://github.com/integritynoble/pwm/tree/main/packages/pwm_core/contrib/modalities/ct_radon.py)

- On-chain L3 spec hash: `pwm_l3_spec:pwm_low_dose_ct_benchmark:1.0.0` on PWM main-net at **[TODO: contract address]**.

Author contributions

[TODO: Fill at submission. Use the CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

[TODO: Declare at submission.]

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